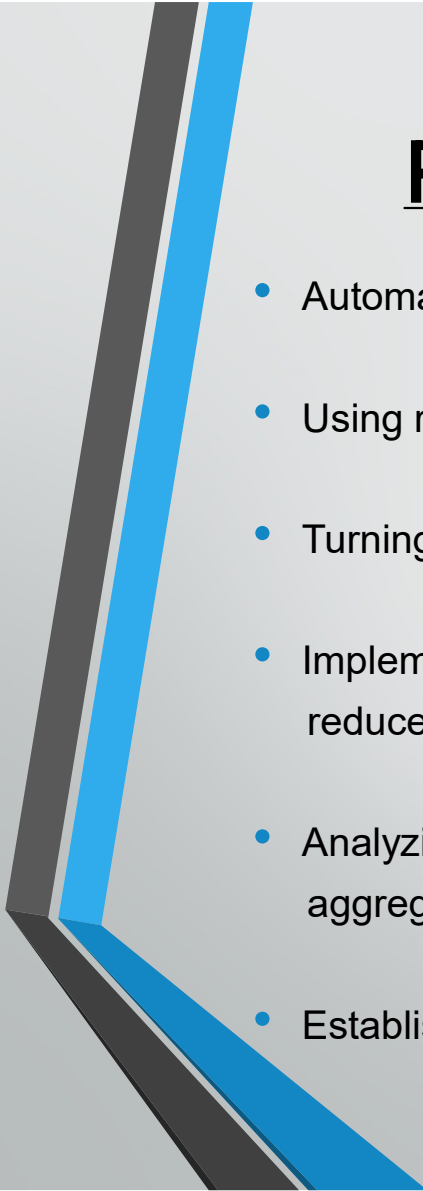


PHARMACY INVENTORY & SALES ANALYTICS SYSTEM

(SQL IMPLEMENTATION)

-BY SIDDHI MORE





PROJECT OVERVIEW & OBJECTIVES

- Automating the tracking of medicine from initial stock entry to final sales.
- Using relational data to categorize medicines and track inventory levels in real-time.
- Turning raw sales data into actionable insights using sql analytics.
- Implementing logic to identify medications nearing expiration to ensure patient safety and reduce wastage.
- Analyzing sales trends and revenue performance using window functions and aggregations.
- Establishing relational constraints between medicines ,inventory and sales.

Relational Data Model

1. Medicines Table

(Stores Core Drug Information)

- Create table Medicines(
Medicine_id number primary key,
Drug_name varchar2(100) not null,
Category varchar2(50),
Unit_type varchar2(20)
);
- Select * from Medicines;

	MEDICINE_ID	DRUG_NAME	CATEGORY	UNIT_TYPE
1	1	Paracetamol 500mg	Analgesic	Tablet
2	2	Amoxicillin 250mg	Antibiotic	Capsule
3	3	Atorvastatin 10mg	Cardiac	Tablet
4	4	Metformin 500mg	Anti-Diabetic	Tablet
5	5	Azithromycin 500mg	Antibiotic	Tablet
6	6	Amlodipine 5mg	Cardiac	Tablet
7	7	Pantoprazole 40 mg	Gastrointestinal	Tablet
8	8	Cetirizine 10mg	Anti-Allergic	Tablet
9	9	Ibuprofen 400mg	Analgesic	Tablet
10	10	Cough Syrup (Dextromethorphan)	Respiratory	Syrup
11	11	Insulin Glargine	Anti-Diabetic	Injection
12	12	Telmisartan 40mg	Cardiac	Tablet
13	13	Multivitamin Gold	Supplement	Capsule
14	14	Diclofenac Gel	Analgesic	Ointment
15	15	Omeprazole 20mg	Gastrointestinal	Capsule
16	16	Losartan 50mg	Cardiac	Tablet
17	17	Gabapentin 300mg	Neuro-psychiatric	Capsule
18	18	Salbutamol Inhaler	Respiratory	Inhaler
19	19	Clopidogrel 75mg	Cardiac	Tablet
20	20	Ranitidine 150mg	Gastrointestinal	Tablet
21	21	Prednisolone 5mg	Steroid	Tablet
22	22	Loperamide 2mg	Gastrointestinal	Capsule
23	23	Fluconazole 150mg	Anti-Fungal	Tablet
24	24	Levocetirizine 5mg	Anti-Allergic	Tablet
25	25	Glimepiride 2mg	Anti-Diabetic	Tablet

2.Inventory Table (Manages Physical Batches And Expiry)

- Create table Inventory(
Batch_ID Varchar(50) primary key,
Medicine_ID number REFERENCES
Medicines (Medicine_ID),
Stock_Quantity number,
Expiry_Date date,
Unit_Price number (10,2)
);
- Select * from Inventory;

	BATCH_ID	MEDICINE_ID	STOCK_QUANTITY	EXPIRY_DATE	UNIT_PRICE
1	BAT-001	1	100	01-12-26	2.5
2	BAT-002	2	15	15-05-26	12
3	BAT-003	3	200	10-01-27	15
4	BAT-004	4	45	20-04-26	8
5	BAT-005	5	10	30-05-26	45
6	BAT-006	6	80	22-03-27	6.5
7	BAT-007	7	120	11-11-26	14
8	BAT-008	8	60	05-08-26	4
9	BAT-009	9	30	15-04-26	5.5
10	BAT-010	10	25	01-06-26	85
11	BAT-011	11	12	10-10-26	450
12	BAT-012	12	90	14-02-27	18
13	BAT-013	13	40	25-12-26	25
14	BAT-014	14	18	01-05-26	65
15	BAT-015	15	55	30-09-26	11
16	BAT-016	16	110	10-05-27	12.5
17	BAT-017	17	45	20-11-26	28
18	BAT-018	18	35	15-08-27	185
19	BAT-019	19	150	12-12-26	14.25
20	BAT-020	20	20	25-04-26	4.5
21	BAT-021	21	65	05-01-27	7
22	BAT-022	22	85	18-07-26	9
23	BAT-023	23	12	10-05-26	55
24	BAT-024	24	95	30-03-27	6
25	BAT-025	25	40	15-10-26	22

3.Sales Table

(Records Daily Sales Activity)

- Create table Sales(
Sale_ID number primary key,
Batch_ID Varchar(50) REFERENCES
Inventory (Batch_ID),
Quantity_Sold number,
Sale_Date date,
Total_Price Number(10,2)
);
- Select * from Sales;

	SALE_ID	BATCH_ID	QUANTITY_SOLD	SALE_DATE	TOTAL_PRICE
1	101	BAT-001	10	30-03-26	25
2	102	BAT-003	2	30-03-26	30
3	103	BAT-009	5	31-03-26	27.5
4	104	BAT-011	1	31-03-26	450
5	105	BAT-001	20	01-04-26	50
6	106	BAT-004	2	01-04-26	16
7	107	BAT-007	10	01-04-26	140
8	108	BAT-010	1	02-04-26	85
9	109	BAT-015	5	02-04-26	55
10	110	BAT-003	5	02-04-26	75
11	111	BAT-006	15	03-04-26	97.5
12	112	BAT-013	4	03-04-26	100
13	113	BAT-001	12	03-04-26	30
14	114	BAT-008	8	03-04-26	32
15	115	BAT-012	3	03-04-26	54
16	116	BAT-016	5	04-04-26	62.5
17	117	BAT-018	1	04-04-26	185
18	118	BAT-020	10	04-04-26	45
19	119	BAT-002	3	04-04-26	36
20	120	BAT-019	20	05-04-26	285
21	121	BAT-025	2	05-04-26	44
22	122	BAT-030	50	05-04-26	275
23	123	BAT-005	1	05-04-26	45
24	124	BAT-021	5	06-04-26	35
25	125	BAT-026	10	06-04-26	350

Inventory & Master Data Management

1) List all medicines that belongs to the 'Antibiotic' category.

- Select * from Medicines where category = 'Antibiotic';

	MEDICINE_ID	DRUG_NAME	CATEGORY	UNIT_TYPE
1	2	Amoxicillin 250mg	Antibiotic	Capsule
2	5	Azithromycin 500mg	Antibiotic	Tablet

2) Find all drugs that are sold as 'Syrup'.

- Select * from Medicines where unit_type = 'Syrup';

	MEDICINE_ID	DRUG_NAME	CATEGORY	UNIT_TYPE
1	10	Cough Syrup (Dextromethorphan)	Respiratory	Syrup

3) Find medicines whose name start with A.

- Select * from Medicines where drug_name like 'A%';

	MEDICINE_ID	DRUG_NAME	CATEGORY	UNIT_TYPE
1	2	Amoxicillin 250mg	Antibiotic	Capsule
2	3	Atorvastatin 10mg	Cardiac	Tablet
3	5	Azithromycin 500mg	Antibiotic	Tablet
4	6	Amlodipine 5mg	Cardiac	Tablet
5	30	Alprazolam 0.25mg	Neuro-psychiatric	Tablet

4) Show all medicines sorted from A to Z by their drug_name.

➤ Select * from Medicines order by drug_name asc;

	MEDICINE_ID	DRUG_NAME	CATEGORY	UNIT_TYPE
1	30	Alprazolam 0.25mg	Neuro-psychiatric	Tablet
2	6	Amlodipine 5mg	Cardiac	Tablet
3	2	Amoxicillin 250mg	Antibiotic	Capsule
4	3	Atorvastatin 10mg	Cardiac	Tablet
5	5	Azithromycin 500mg	Antibiotic	Tablet
6	8	Cetirizine 10mg	Anti-Allergic	Tablet
7	19	Clopidogrel 75mg	Cardiac	Tablet
8	10	Cough Syrup (Dextromethorphan)	Respiratory	Syrup
9	14	Diclofenac Gel	Analgesic	Ointment
10	29	Enoxaparin 40mg	Anticoagulant	Injection
11	23	Fluconazole 150mg	Anti-Fungal	Tablet
12	17	Gabapentin 300mg	Neuro-psychiatric	Capsule
13	25	Glimepiride 2mg	Anti-Diabetic	Tablet
14	9	Ibuprofen 400mg	Analgesic	Tablet
15	11	Insulin Glargine	Anti-Diabetic	Injection
16	24	Levocetirizine 5mg	Anti-Allergic	Tablet
17	22	Loperamide 2mg	Gastrointestinal	Capsule
18	16	Losartan 50mg	Cardiac	Tablet
19	4	Metformin 500mg	Anti-Diabetic	Tablet
20	13	Multivitamin Gold	Supplement	Capsule
21	28	Mupirocin Ointment	Antibacterial	Ointment
22	15	Omeprazole 20mg	Gastrointestinal	Capsule
23	27	Ondansetron 4mg	Antimetic	Tablet
24	7	Pantoprazole 40 mg	Gastrointestinal	Tablet
25	1	Paracetamol 500mg	Analgesic	Tablet

5) Search for medicines that contain specific numerical strength such as 500mg or 650mg anywhere in their name.

- Select drug_name ,category from Medicines where REGEXP_LIKE(drug_name,'500|650');

	DRUG_NAME	CATEGORY
1	Paracetamol 500mg	Analgesic
2	Metformin 500mg	Anti-Diabetic
3	Azithromycin 500mg	Antibiotic

6) List all distinct therapeutic categories currently in the database.

- Select distinct category from Medicines;

	CATEGORY
1	Analgesic
2	Antibiotic
3	Cardiac
4	Anti-Diabetic
5	Gastrointestinal
6	Anti-Allergic
7	Respiratory
8	Supplement
9	Neuro-psychiatric
10	Steroid
11	Anti-Fungal
12	Antimetic
13	Antibacterial
14	Anticoagulant

Operational Risk & Stock Control

1) Find the batch_id and expiry_date for 'Paracetamol500mg'.

➤ Select m.drug_name,i.batch_id, i.expiry_date from Medicines m inner join Inventory I on m.medicine_id = i.medicine_id where m.drug_name = 'Paracetamol500mg';

	DRUG_NAME	BATCH_ID	EXPIRY_DATE
1	Paracetamol 500mg	BAT-001	01-12-26

2) Find all batches expiring in the next 30 days.

➤ Select * from Inventory where expiry_date between TO_DATE('2026-04-01','YYYY-MM-DD') and TO_DATE('2026-04-30','YYYY-MM-DD');

	BATCH_ID	MEDICINE_ID	STOCK_QUANTITY	EXPIRY_DATE	UNIT_PRICE
1	BAT-004	4	45	20-04-26	8
2	BAT-009	9	30	15-04-26	5.5
3	BAT-020	20	20	25-04-26	4.5

3) List drugs where the quantity is less than 20 units.

➤ Select * from Inventory where stock_quantity < 20;

	BATCH_ID	MEDICINE_ID	STOCK_QUANTITY	EXPIRY_DATE	UNIT_PRICE
1	BAT-002	2	15	15-05-26	12
2	BAT-005	5	10	30-05-26	45
3	BAT-011	11	12	10-10-26	450
4	BAT-014	14	18	01-05-26	65
5	BAT-023	23	12	10-05-26	55
6	BAT-028	28	15	15-06-26	110
7	BAT-029	29	8	05-09-26	620

4) Find all batch_ids and medicine_id that are set to expire before june 2026.

➤ Select batch_id,medicine_id ,expiry_date from Inventory where expiry_date < TO_DATE('2026-06-01','YYYY-MM-DD');

	BATCH_ID	MEDICINE_ID	EXPIRY_DATE
1	BAT-002	2	15-05-26
2	BAT-004	4	20-04-26
3	BAT-005	5	30-05-26
4	BAT-009	9	15-04-26
5	BAT-014	14	01-05-26
6	BAT-020	20	25-04-26
7	BAT-023	23	10-05-26

5) List batches sorted by their expiry_date(oldest first).

➤ Select * from Inventory order by expiry_date asc;

	BATCH_ID	MEDICINE_ID	STOCK_QUANTITY	EXPIRY_DATE	UNIT_PRICE
1	BAT-009	9	30	15-04-26	5.5
2	BAT-004	4	45	20-04-26	8
3	BAT-020	20	20	25-04-26	4.5
4	BAT-014	14	18	01-05-26	65
5	BAT-023	23	12	10-05-26	55
6	BAT-002	2	15	15-05-26	12
7	BAT-005	5	10	30-05-26	45
8	BAT-010	10	25	01-06-26	85
9	BAT-028	28	15	15-06-26	110
10	BAT-022	22	85	18-07-26	9
11	BAT-008	8	60	05-08-26	4
12	BAT-027	27	50	22-08-26	16
13	BAT-029	29	8	05-09-26	620
14	BAT-015	15	55	30-09-26	11
15	BAT-011	11	12	10-10-26	450
16	BAT-025	25	40	15-10-26	22
17	BAT-007	7	120	11-11-26	14
18	BAT-017	17	45	20-11-26	28
19	BAT-001	1	100	01-12-26	2.5
20	BAT-019	19	150	12-12-26	14.25
21	BAT-013	13	40	25-12-26	25
22	BAT-021	21	65	05-01-27	7
23	BAT-003	3	200	10-01-27	15
24	BAT-030	30	120	10-02-27	5.5
25	BAT-012	12	90	14-02-27	18

6) Find drugs that are both low stock and expiring soon.

➤ Select m.drug_name from Medicines m inner join Inventory i on m.medicine_id = i.medicine_id where i.stock_quantity < 20 and i.expiry_date < to_date('2026-06-01','YYYY-MM-DD');

	DRUG_NAME
1	Amoxicillin 250mg
2	Azithromycin 500mg
3	Diclofenac Gel
4	Fluconazole 150mg

3) List batchids sorted by the total revenue they generated.

- Select batch_id,sum(total_price) from Sales group by batch_id order by sum(total_price) desc;

	BATCH_ID	SUM(TOTAL_PRICE)
1	BAT-029	620
2	BAT-011	450
3	BAT-026	350
4	BAT-019	285
5	BAT-030	275
6	BAT-018	185
7	BAT-007	140
8	BAT-014	130
9	BAT-017	112
10	BAT-001	105
11	BAT-003	105
12	BAT-013	100
13	BAT-006	97.5
14	BAT-024	90
15	BAT-010	85
16	BAT-022	72
17	BAT-016	62.5
18	BAT-015	55
19	BAT-012	54
20	BAT-005	45
21	BAT-020	45
22	BAT-025	44
23	BAT-002	36
24	BAT-021	35
25	BAT-008	32

4) Calculate the total potential revenue if all current stock was sold.

- Select sum(stock_quantity * unit_price) as total_potential_revenue from Inventory;

	TOTAL_POTENTIAL_REVENUE
1	48502.5

5) Category wise revenue with rank.

- Select m.category,sum(s.Total_price) as revenue,rank() over (order by sum(s.total_price)desc) as rank from Sales s join Inventory i on s.batch_id = i.batch_id join medicines m on i.medicine_id = m.medicine_id group by m.category;

⚡ CATEGORY	⚡ REVENUE	⚡ RANK
1 Anticoagulant	620	1
2 Cardiac	604	2
3 Anti-Diabetic	510	3
4 Supplement	450	4
5 Neuro-psychiatric	387	5
6 Gastrointestinal	312	6
7 Respiratory	270	7
8 Analgesic	262.5	8
9 Anti-Allergic	122	9
10 Antibiotic	81	10
11 Steroid	35	11

6) Find Min, Max and Avg sale.

- Select max(quantity_sold) as highest_sale from Sales;
- Select min(quantity_sold)as lowest_sale from Sales;
- Select avg(quantity_sold) as average_sale from Sales;

⚡ HIGHEST_SALE
1 50

⚡ LOWEST_SALE
1 1

⚡ AVERAGE_SALE
1 8

7) How many different medicines do we have in each category.

- Select category,count(*) as medicine_count from Medicines group by category;

⚡	CATEGORY	⚡	MEDICINE_COUNT
1	Analgesic		3
2	Antibiotic		2
3	Cardiac		5
4	Anti-Diabetic		3
5	Gastrointestinal		4
6	Anti-Allergic		2
7	Respiratory		2
8	Supplement		2
9	Neuro-psychiatric		2
10	Steroid		1
11	Anti-Fungal		1
12	Antimetic		1
13	Antibacterial		1
14	Anticoagulant		1

8) List all sales where the quantity_sold was more than 5 and the total_price was more than Rs.100.

- Select * from Sales where quantity_sold > 5 and total_price > 100;

⚡	SALE_ID	⚡	BATCH_ID	⚡	QUANTITY_SOLD	⚡	SALE_DATE	⚡	TOTAL_PRICE
1	107	BAT-007		10	01-04-26		140		
2	120	BAT-019		20	05-04-26		285		
3	122	BAT-030		50	05-04-26		275		
4	125	BAT-026		10	06-04-26		350		

9) Find the most expensive medicine in the stock based on unit_price.

- Select * from Inventory where unit_price = (select max(unit_price) from Inventory);

	BATCH_ID	MEDICINE_ID	STOCK_QUANTITY	EXPIRY_DATE	UNIT_PRICE
1	BAT-029	29	8	05-09-26	620

10) Calculate the total number of units available for 'Metformin' across all batches.

- Select sum(stock_quantity)from Inventory where medicine_id=4;

	SUM(STOCK_QUANTITY)
1	45

Advanced Analytical Techniques

1) List all medicines and their inventory (including those not in stock).

➤ Select m.drug_name,i.batch_id,
i.stock_quantity from Medicines m left join
Inventory i on m.medicine_id =
i.medicine_id;

	DRUG_NAME	BATCH_ID	STOCK_QUANTITY
1	Paracetamol 500mg	BAT-001	100
2	Amoxicillin 250mg	BAT-002	15
3	Atorvastatin 10mg	BAT-003	200
4	Metformin 500mg	BAT-004	45
5	Azithromycin 500mg	BAT-005	10
6	Amlodipine 5mg	BAT-006	80
7	Pantoprazole 40 mg	BAT-007	120
8	Cetirizine 10mg	BAT-008	60
9	Ibuprofen 400mg	BAT-009	30
10	Cough Syrup (Dextromethorphan)	BAT-010	25
11	Insulin Glargine	BAT-011	12
12	Telmisartan 40mg	BAT-012	90
13	Multivitamin Gold	BAT-013	40
14	Diclofenac Gel	BAT-014	18
15	Omeprazole 20mg	BAT-015	55
16	Losartan 50mg	BAT-016	110
17	Gabapentin 300mg	BAT-017	45
18	Salbutamol Inhaler	BAT-018	35
19	Clopidogrel 75mg	BAT-019	150
20	Ranitidine 150mg	BAT-020	20
21	Prednisolone 5mg	BAT-021	65
22	Loperamide 2mg	BAT-022	85
23	Fluconazole 150mg	BAT-023	12
24	Levocetirizine 5mg	BAT-024	95
25	Glimepiride 2mg	BAT-025	40

2) Find sales for batches that might have been removed from Inventory.

➤ Select s.sale_id, s.batch_id, i.expiry_date
from Inventory i right join Sales s on
i.batch_id = s.batch_id;

	SALE_ID	BATCH_ID	EXPIRY_DATE
1	101	BAT-001	01-12-26
2	105	BAT-001	01-12-26
3	113	BAT-001	01-12-26
4	119	BAT-002	15-05-26
5	102	BAT-003	10-01-27
6	110	BAT-003	10-01-27
7	106	BAT-004	20-04-26
8	123	BAT-005	30-05-26
9	111	BAT-006	22-03-27
10	107	BAT-007	11-11-26
11	114	BAT-008	05-08-26
12	103	BAT-009	15-04-26
13	108	BAT-010	01-06-26
14	104	BAT-011	10-10-26
15	115	BAT-012	14-02-27
16	112	BAT-013	25-12-26
17	127	BAT-014	01-05-26
18	109	BAT-015	30-09-26
19	116	BAT-016	10-05-27
20	128	BAT-017	20-11-26
21	117	BAT-018	15-08-27
22	120	BAT-019	12-12-26
23	118	BAT-020	25-04-26
24	124	BAT-021	05-01-27
25	126	BAT-022	18-07-26

3) Find medicines in the same category.

➤ Select a.drug_name as medicine_A, b.drug_name as medicine_B, a.category from Medicines a join Medicines b on a.category = b.category where a.medicine_id < b.medicine_id;

	MEDICINE_A	MEDICINE_B	CATEGORY
1	Amoxicillin 250mg	Azithromycin 500mg	Antibiotic
2	Atorvastatin 10mg	Amlodipine 5mg	Cardiac
3	Paracetamol 500mg	Ibuprofen 400mg	Analgesic
4	Metformin 500mg	Insulin Glargine	Anti-Diabetic
5	Atorvastatin 10mg	Telmisartan 40mg	Cardiac
6	Amlodipine 5mg	Telmisartan 40mg	Cardiac
7	Paracetamol 500mg	Diclofenac Gel	Analgesic
8	Ibuprofen 400mg	Diclofenac Gel	Analgesic
9	Pantoprazole 40 mg	Omeprazole 20mg	Gastrointestinal
10	Atorvastatin 10mg	Losartan 50mg	Cardiac
11	Amlodipine 5mg	Losartan 50mg	Cardiac
12	Telmisartan 40mg	Losartan 50mg	Cardiac
13	Cough Syrup(Dextromethorphan)	Salbutamol Inhaler	Respiratory
14	Atorvastatin 10mg	Clopidogrel 75mg	Cardiac
15	Amlodipine 5mg	Clopidogrel 75mg	Cardiac
16	Telmisartan 40mg	Clopidogrel 75mg	Cardiac
17	Losartan 50mg	Clopidogrel 75mg	Cardiac
18	Pantoprazole 40 mg	Ranitidine 150mg	Gastrointestinal
19	Omeprazole 20mg	Ranitidine 150mg	Gastrointestinal
20	Pantoprazole 40 mg	Loperamide 2mg	Gastrointestinal
21	Omeprazole 20mg	Loperamide 2mg	Gastrointestinal
22	Ranitidine 150mg	Loperamide 2mg	Gastrointestinal
23	Cetirizine 10mg	Levocetirizine 5mg	Anti-Allergic
24	Metformin 500mg	Glimepiride 2mg	Anti-Diabetic
25	Insulin Glargine	Glimepiride 2mg	Anti-Diabetic

4) Generate a master checklist that pairs every medicine name with every existing batch number for a manual warehouse audit.

➤ Select m.drug_name ,i.batch_id
from Medicines m cross join
Inventory i;

DRUG_NAME	BATCH_ID
1 Paracetamol 500mg	BAT-001
2 Amoxicillin 250mg	BAT-001
3 Atorvastatin 10mg	BAT-001
4 Metformin 500mg	BAT-001
5 Azithromycin 500mg	BAT-001
6 Amlodipine 5mg	BAT-001
7 Pantoprazole 40 mg	BAT-001
8 Cetirizine 10mg	BAT-001
9 Ibuprofen 400mg	BAT-001
10 Cough Syrup (Dextromethorphan)	BAT-001
11 Insulin Glargine	BAT-001
12 Telmisartan 40mg	BAT-001
13 Multivitamin Gold	BAT-001
14 Diclofenac Gel	BAT-001
15 Omeprazole 20mg	BAT-001
16 Losartan 50mg	BAT-001
17 Gabapentin 300mg	BAT-001
18 Salbutamol Inhaler	BAT-001
19 Clopidogrel 75mg	BAT-001
20 Ranitidine 150mg	BAT-001
21 Prednisolone 5mg	BAT-001
22 Loperamide 2mg	BAT-001
23 Fluconazole 150mg	BAT-001
24 Levocetirizine 5mg	BAT-001
25 Glimepiride 2mg	BAT-001

DRUG_NAME	BATCH_ID
26 Vitamin D3 60K	BAT-001
27 Ondansetron 4mg	BAT-001
28 Mupirocin Ointment	BAT-001
29 Enoxaparin 40mg	BAT-001
30 Alprazolam 0.25mg	BAT-001
31 Paracetamol 500mg	BAT-002
32 Amoxicillin 250mg	BAT-002
33 Atorvastatin 10mg	BAT-002
34 Metformin 500mg	BAT-002
35 Azithromycin 500mg	BAT-002
36 Amlodipine 5mg	BAT-002
37 Pantoprazole 40 mg	BAT-002
38 Cetirizine 10mg	BAT-002
39 Ibuprofen 400mg	BAT-002
40 Cough Syrup (Dextromethorphan)	BAT-002
41 Insulin Glargine	BAT-002
42 Telmisartan 40mg	BAT-002
43 Multivitamin Gold	BAT-002
44 Diclofenac Gel	BAT-002
45 Omeprazole 20mg	BAT-002
46 Losartan 50mg	BAT-002
47 Gabapentin 300mg	BAT-002
48 Salbutamol Inhaler	BAT-002
49 Clopidogrel 75mg	BAT-002

Advanced Database Objects

- Create a view to monitor medicines that fall into specific mid tier price range.
- Create view medicine_mid_price as select m.drug_name,m.category,i.unit_price from Medicines m inner join Inventory i on m.medicine_id = i.medicine_id where i.unit_price > 100 and i.unit_price < 500;
- Select * from Medicine_mid_price;

	DRUG_NAME	CATEGORY	UNIT_PRICE
1	Insulin Glargine	Anti-Diabetic	450
2	Salbutamol Inhaler	Respiratory	185
3	Mupirocin Ointment	Antibacterial	110

Sales Analysis via Subqueries

1) Find inventory for all medicines in the 'Anti-Allergic' category.

- Select batch_id, stock_quantity, expiry_date from Inventory where medicine_id in (Select medicine_id from Medicines where category ='Anti-Allergic');

	BATCH_ID	STOCK_QUANTITY	EXPIRY_DATE
1	BAT-008	60	05-08-26
2	BAT-024	95	30-03-27

2) List all sales for 'Alprazolam 0.25mg'

➤ Select sale_id, quantity_sold, total_price
from Sales where batch_id in (select
batch_id from Inventory where
medicine_id =(select medicine_id from
Medicines where drug_name
='Alprazolam 0.25mg'));

	SALE_ID	QUANTITY_SOLD	TOTAL_PRICE
1	122	50	275

Conclusion

- Successfully Integrated Pharmacy Inventory And sales into relational database.
- Created automated alerts for low stock and expiring batches to ensure 100% patient safety.
- Provided management with ranking and revenue insights to identify high performing medicine categories.
- Used advanced joins to connect Sales, Inventory and Master records for 360 degree reporting.
- These techniques demonstrate the ability to handle complex ,real-world data.

THANKYOU!

Siddhi more

Pharmacist & Data Analytics Aspirant